

# Transition Analysis Toolkit

## A Reproducible Analysis of Student Transition Pathways

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# 1 Introduction

## 1.1 Research Questions

1. Which factors are the strongest predictors of success in first-year university mathematics?
2. How consistent are predictor importance rankings across multiple machine learning models?
3. What educational insights can be drawn from the tuned models?

## 2 Methodology

## 3 Results

### 3.1 Model Performance

Table 1: ROC AUC comparison for cross-validated Decision Tree and Random Forest models.

| Model                        | CV ROC AUC | Test ROC AUC | Test Rows |
|------------------------------|------------|--------------|-----------|
| Decision Tree — With Risk    | 0.7072     | 0.7664       | 295       |
| Random Forest — Without Risk | 0.7101     | 0.7603       | 295       |
| Random Forest — With Risk    | 0.7095     | 0.7599       | 295       |
| Decision Tree — Without Risk | 0.6994     | 0.7557       | 295       |

### 3.2 Feature Importance

| Rank | Research theme                    | Mean importance |
|------|-----------------------------------|-----------------|
| 1    | Secondary mathematics preparation | 0.2755          |
| 2    | Composite risk measure            | 0.2381          |
| 3    | Preparatory mathematics           | 0.2320          |
| 4    | Previous university performance   | 0.1090          |
| 5    | Secondary school achievement      | 0.1058          |
| 6    | Student characteristics           | 0.0395          |

### 3.3 Feature Rankings

| Rank | Feature                                  | Models | Mean importance |
|------|------------------------------------------|--------|-----------------|
| 1    | Composite risk index                     | 2      | 0.4762          |
| 2    | Prior preparatory mathematics            | 4      | 0.1694          |
| 3    | Secondary mathematics: Other             | 4      | 0.1019          |
| 4    | Secondary mathematics: Standard          | 4      | 0.0950          |
| 5    | Previous attempt at current subject      | 4      | 0.0693          |
| 6    | Previous preparatory mathematics pass    | 4      | 0.0532          |
| 7    | Previous failure in current subject      | 4      | 0.0397          |
| 8    | International student                    | 4      | 0.0395          |
| 9    | Secondary school mark below 70           | 4      | 0.0394          |
| 10   | Secondary mathematics: Extension 1       | 4      | 0.0348          |
| 11   | Secondary school mark 90 or above        | 4      | 0.0288          |
| 12   | Secondary mathematics: Extension 2       | 4      | 0.0257          |
| 13   | Secondary school mark 80–89              | 4      | 0.0228          |
| 14   | Secondary mathematics: Advanced          | 4      | 0.0182          |
| 15   | Secondary school mark 70–79              | 4      | 0.0148          |
| 16   | Previous preparatory mathematics failure | 4      | 0.0094          |

### 3.4 Model Interpretation

Across the tuned tree-based models, secondary mathematics preparation emerged as the most influential predictor group (mean importance = 0.276). This was followed by the composite risk measure (0.238) and preparatory mathematics (0.232). Together, these three groups accounted for 74.6% of the average feature importance across the tuned models. Secondary mathematics preparation remained influential across all four models, whereas the composite risk measure contributed strongly but appeared in two of the four models. In contrast, student characteristics made the smallest average contribution. This suggests that prior secondary mathematics preparation was the strongest and most consistent predictor of success across the tuned models.

## 4 Discussion

## 5 Conclusion

## References

## A Supplementary Material